

User Manual — Tournament Scheduler (short)

This short manual shows how to use the GUI and run experiments.

1) Start the application

- Create and activate a virtual environment (optional but recommended):

```
```powershell
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
```
```

- Run the GUI:

```
```powershell
python src/src/main.py
```
```

2) GUI quick actions

- Enter `Number of teams` (e.g., 6, 8, 10). If odd, a bye will be added automatically.
- Choose `Algorithm`: Baseline (Random), Greedy (Heuristic), or Genetic Algorithm.
- Adjust GA parameters (Population, Generations, Crossover rate, Mutation rate) when using GA.
- Click `Run Selected` to perform a single run and view the schedule and conflict counts.
- Click `Run Both (Compare)` to execute the greedy baseline and three GA presets and view comparisons.
- Use `Load Teams` to load team names (one per line) — the schedule display will show names.
- Use `Load Venues` to load venue names for display only.
- Use `Show Distribution` or `Open Graph` to visualize results (requires `matplotlib`).

3) Running batch experiments

- Use `src/experiments/run\_experiments.py` to run the preconfigured experiments and save results to `src/experiments\_output/`.

4) Exporting results

- Use `Save Results` in the GUI to export the last run to a text file.

Notes

- If `matplotlib` is not installed, plotting features will be disabled.
- Convert this `UserManual.md` to PDF for submission if a PDF is required (e.g., using a Markdown-to-PDF tool).